

Jay Patil

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Education

Rutgers University, New Jersey – *Master of Science in Computer Science*

Sept 2022 - May 2024

COURSES: Algorithms, Machine Learning, Distributed Systems, OS, DBMS

CGPA: 3.71/4.00

Pune Institute of Computer Technology, Pune – *Bachelor of Engineering in Computer Engineering*

Aug 2016 - June 2020

COURSES: Advanced Data Structures, Computer Networks, Computer Architecture

CGPA: 8.97/10.00

Experience

Antonina Mitrofanova Lab – Research Assistant, Software Developer

Sept 2022 – May 2024

- Designed and developed a **user-engaging and cross-browser compatible** web application for data analysis and rendering of complex human gene-splicing networks consisting of more than 2k nodes using Angular, and Cytoscape.js.
- Designed **ETL pipelines** to compile and cleanse data from sources like NCBI and making it ready for Cox Analysis.
- Conducted **Principal Component Analysis** on human gene dataset and reduced dimensionality to improve compute time of training and eliminate unnecessary correlated variable.

Veritas Technologies LLC – Software Engineer, Full Stack

Sept 2020 – Jul 2022

- Worked in a **Scrum-based agile** environment and developed intuitive user interfaces using Angular to facilitate the data discovery of more than **1 billion** messages and emails from major data sources like Teams, Outlook, Slack, etc.
- Took the ownership to update old code base to modern development standards making it **more secure** by implementing API authorization in .NET Core and **reducing loading time** of the application by **46%** by implementing lazy loading of pages.
- Leveraged **Elasticsearch** to implement fuzzy search on more than a billion MS Teams messages to retrieve relevant data.
- Developed a web service to integrate and connect One Drive services from Merge1 to Veritas' internal platform.
- Implemented Azure Storage REST APIs using C# in .NET to retrieve data from Blob storage to support the email export service.
- Optimized stored procedures in MSSQL by using **indexed columns** in queries and writing modular code.
- Coordinated efficient large-scale features deployments using Terraform and Puppet and managed CI/CD using TFS.

Projects

Distributed Data Store Optimization | Linux, C, dqlite, Raft

- Implemented a query batching and pipelining optimizations to a distributed highly available key value store(dqlite) built using sqlite and raft consensus protocol (open-source canonical implementation) achieving **10% reduction in response time**. Created benchmarks to measure performance of different scenarios like node failure, membership changes, network partition.

Attention Mechanisms in Image Captioning | Python, Linux, PyTorch, NumPy, Pandas

- Led a comprehensive comparative analysis of RATT, XLAN, and Dual Attention models in Computer Vision. Utilized TensorFlow and PyTorch frameworks to meticulously implement and fine-tune the models. Employed the METEOR metric to assess the quality of complex texts, providing nuanced insights into the strengths and potential areas for enhancement of each model.

Text-to-Video Generation through Advanced Diffusion Models (GenAI) | Python, Linux, TensorFlow

- Spearheaded a pioneering text-to-video (T2V) generation project utilizing the "Tune-A-Video" methodology, focused on **one-shot tuning** of image diffusion models to elevate the quality and contextual accuracy of video output across diverse data domains. Utilized TensorFlow and PyTorch frameworks to meticulously implement and fine-tune model.

Skills

Languages: Go, Python3, C#, Java, JavaScript, SQL, C/C++, HTML/CSS, TypeScript

Web Technologies: .NET, SpringBoot, NodeJS, React, Express, Angular, Flask, Web3, O'Auth, FastAPI, Redux

Tools and Frameworks: LangChain, Kafka, Azure, Docker, AWS, Redis, DynamoDB, Git, MongoDB, Kubernetes, Map-Reduce

Awards & Recognition

- Award for 'Team Collaboration and Inclusion' for collaboration with teams across the globe at Veritas.